Non-Invasive Leakage Severity Estimation in Water Distribution Networks using Wavelet Scattering Transforms: A Comparative Sensor Study

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***Abstract*—The management of water loss is a serious issue on the international arena where existing data-driven methods are mainly based on the binary leakage or leak type classification without considering the quantification of the severity of the leakage (flow magnitude). This paper introduces a new model of estimating the severity of leakage using the Wavelet Scattering Transform (WST) and Support Vector Machines (SVM). We tested the abilities of three specific sensor modalities, namely hydrophones (invasive), accelerometers (non-invasive) and dynamic pressure sensors (invasive) on a branched water distribution network testbed under three hydraulic states, which are No Demand, Low Severity (0.18 L/s), and High Severity (0.47 L/s). Each 30-second recording was segmented into 1-second windows, yielding 510 Accelerometer segments, 541 Pressure Sensor segments, and 900 Hydrophone segments. Under 5-Fold Stratified Cross-Validation, the non-invasive Accelerometer achieved 99.0% $\pm$ 1.2% mean classification accuracy compared to 93.7% $\pm$ 1.9% for the invasive Pressure Sensor and 42.0% $\pm$ 1.7% for the Hydrophone. Linear Discriminant Analysis (LDA) visualization in feature space confirms well-separated clusters for the Accelerometer, while invasive sensors exhibit overlapping distributions attributable to signal saturation and acoustic ambiguity at high flow rates. These results indicate that the use of WST-based models that employ inexpensive, external accelerometers could be useful as a Virtual Flow Meter proxy to help utilities have a scalable means of maintenance prioritization under the evaluated testbed conditions.**

***Index Terms*—Leak detection, Wavelet Scattering Transform, Severity Estimation, Accelerometer, Water Distribution Networks.**

I. Introduction

Loss of water in distribution networks (WDNs) is a significant problem facing the world, with Non-Revenue Water (NRW) consuming up to half of the municipal supply, at a high economic and infrastructure expense. Binary leak detection is well-investigated, but the estimation of the severity (flow magnitude) of leak is an open operational issue. To rank maintenance operations, utilities need to have a proper estimation of severity so that the most important ones can be fixed first since the high volume and critical leaks are the ones that should be fixed first.

Traditional leak detectors tend to take the form of invasive sensors, i.e., hydrophones or dynamic pressure sensors, which are used to measure acoustic waves or pressure transients in the fluid column. Although effective, these sensors are complex and costly to install and vulnerable to bio-fouling and signal saturation in high-turbulence. On the other hand, non-invasive sensors can be mounted, e.g., accelerometers externally on the pipe surface, which is a cost-effective and scalable alternative.

Other recent papers have investigated other data-driven methods of leak monitoring. As an example, Ismail et al. [3] and Butterfield et al. [5] showed that the vibrations caused by the leak could be successfully measured using accelerometers, but in their experiment, the emphasis was made on binary detection (Leak vs No Leak) instead of severity quantification. Mashford et al. [4] also, on the same note, investigated severity quantification.

The success of [4] is that Support Vector Machines (SVM) was effectively used on vibration data with high accuracy in controlled environments. Nevertheless, these legacy techniques tend to be based on manual engineered characteristics (e.g. FFT peak analysis), and do not capture the non-stationary dynamics of transient leaks. More recently, Xu et al. [7] proposed the Wavelet Scattering Transform (WST) in the context of pipeline monitoring, and they pointed to its capability of producing stable and translation-invariant representations. Based on these premises, this paper will specifically aim at filling the gap of severity estimation through the union of the strong feature extraction of WST and the computational speed of SVMs.

Unlike existing studies that apply WST or vibration-based methods primarily to binary leak detection or leak-type classification, this paper focuses on severity estimation across three hydraulic states. In addition, the study compares three sensor modalities (Accelerometer, Pressure Sensor, Hydrophone) under the same testbed conditions to evaluate their effectiveness for severity classification. The combination of WST feature extraction with an SVM classifier provides a lightweight approach that does not require a large labeled dataset or GPU-based computation.

This paper has summarized its main contributions as follows:

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- The comparison of three sensor modalities (Accelerometer, Pressure Sensor, Hydrophone) in classifying the severity.
- A classification system achieving 99.0% $\pm$ 1.2% mean accuracy under 5-Fold Stratified Cross-Validation for distinguishing No Demand, Low Severity (0.18 L/s), and High Severity (0.47 L/s) states.
- Empirical data which show that non-invasive structural vibration is a more predictable indicator of flow magnitude than invasive fluid pressure techniques.

II. RELATED WORK

The leakage detection issue in WDNs has matured from manual acoustic surveys to automated data-driven methods. The initial methods were mainly based on acoustic listening devices and correlators. Nonetheless, as mentioned by Martinez-Rios et al. [1], these are labor-intensive techniques which are highly vulnerable to environmental noise, especially in a city environment where traffic vibrations and industrial vibrations can mask leakage signatures. According to recent surveys [6], although hydrophones are sensitive, they tend to saturate when detecting leakages in near-field conditions, which is overcome by vibration-based detectors.

A. Pressure-Based Methods

The pressure transient analysis is used to examine negative pressure waves caused by the rupture of the pipes. Although these systems are effective in large bursts, they have problems with incipient cracks that are associated with minor drops in pressure. Moreover, dynamic pressure sensors are invasive, which is expensive to install and maintain as they are subject to bio-fouling and drift.

Vibration and Acceleration

The latest research has been redirected to non-invasive monitoring of vibration. Structural energy transfer by flow turbulence can be detected by accelerometers mounted on the outer wall of the pipe. In comparison to hydrophones, which detect sound carried by fluids, accelerometers detect the vibration of the shell of a pipe, which is a mechanical filter, and inherently filters out some of the high-frequency ambient noise. Nonetheless, it is difficult to extract credible features from raw vibration data. Conventional methods make use of Fast Fourier Transform (FFT) characteristics; however, FFT presumes signal stationarity, which is frequently not true in turbulent flow owing to its transient characteristics.

Machine Learning in WDNs

Such classifiers as Support Vector Machines (SVM) and Convolutional Neural Networks (CNNs) have potential as automated detectors [9]. Although CNNs are used to automate feature engineering, they need large datasets and considerable computing resources. The current research addresses these gaps through the Wavelet Scattering Transform (WST), which offers discriminative capability without a substantial amount of training data, coupled with a lightweight SVM.

III. METHODOLOGY

A. Dataset and Sensor Specifications

The data used in the study were taken on the experimental testbed described in [1]. As depicted by the hierarchical form of the dataset, the uncoded time-series measurements are arranged in a systematic way into numerous categories based on several dimensions: sensor modality, network topology, leak type, and hydraulic state.

The data covers three major modalities (Accelerometers, Dynamic Pressure Sensors, Hydrophones) and it is further categorized according to network topology (Branched and Looped). Extensive recording of the structural conditions is done including the base conditions of No-leak and particular types of faults such as Circumferential Crack, Gasket Leak, Longitudinal Crack, and Orifice Leak. The individual data files are contained in CSV format and a structured naming convention is used to file them that indicates these parameters. The dataset records 3 distinct hydraulic conditions, namely a control condition (No Demand (ND)) state, Low Severity leak (0.18 L/s), and High Severity leak (0.47 L/s). Continuous 30-second recordings were made in each state of hydraulics.

After segmentation, the final dataset comprised 510 Accelerometer segments, 541 Dynamic Pressure Sensor segments, and 900 Hydrophone segments. The Accelerometer dataset contained 170 segments for each of the three severity classes.

In order to determine sensor efficacy in the various physical domains, data was measured simultaneously with three different sensor modalities at strategic locations on the network:

- Accelerometer (Non-Invasive): a PCB 333B50 sensor (1000 mV/g sensitivity) that would be mounted to the pipe externally to detect structural vibrations.
- Dynamic Pressure Sensor (Invasive): This sensor is a PCB 102B16 sensor (50 mV/psi sensitivity) mounted in direct contact with the fluid column, used to detect transient pressure drops.
- Hydrophone (Invasive): A -180 dB sensitive Aquarian H2a hydrophone submerged in water to record acoustic emissions in the fluid.

TABLE I
EXPERIMENTAL INSTRUMENTATION SPECIFICATIONS

Component	Specification	Value
Pipe Network	Material Diameter Topology	HDPE 25 mm Branched
Accelerometer	Model Sensitivity	PCB 333B50 1000 mV/g
Pressure Sensor	Model Sensitivity	PCB 102B16 50 mV/psi
Hydrophone	Model Sensitivity	Aquarian H2a -180 dB
DAQ System	Sampling Rate (Vib/Press) Sampling Rate (Hydro)	51.2 kHz 8.0 kHz

The instrumentation that was chosen was well-selected to record the high-frequency transients of turbulent leaks as discussed in Table I. The sampling rate of the DAQ system (51.2 kHz of both the Accelerometer and pressure sensor) is especially important in the resolution of the thin structural vibration modes of the very high frequencies that are necessary to the successful extraction of the Wavelet Scattering Transform feature.

B. Proposed Framework Architecture

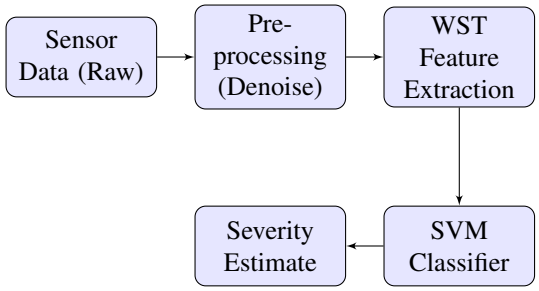


Fig. 1. Block Diagram of the proposed Non-Invasive Leakage Severity Estimation Framework. The WST acts as a non-linear demodulator to extract stable features from raw sensor data.

The general process flow of the suggested severity estimation system is shown in Fig. 1. It has four stages, including data acquisition, signal preprocessing, wavelet scattering feature extraction, and SVM classification. The proposed framework algorithmic procedure is formally outlined in Algorithm 1.

C. Data Preprocessing and Segmentation

The raw data obtained in the testbed were a continuous 30 seconds record of each hydraulic state. We used non-overlapping segmentation strategy to create enough training samples to use in the machine learning model. The 30-second samples were cut into 1-second samples to give 30 independent samples per recording.

Each segment was mean-centred to remove the DC offset. Wavelet denoising was then applied using a four-level Symlet-3 (sym3) decomposition. The noise level was estimated from the finest-scale detail coefficients using the median absolute deviation:

$$\hat{\sigma} = \frac{\text{median}(|c_{-1}|)}{0.6745}, \qquad \lambda = \hat{\sigma}\sqrt{2\ln N} \tag{1}$$

where N is the segment length. Soft thresholding was applied to the wavelet coefficients followed by signal reconstruction. This preprocessing step reduces broadband noise while preserving the transient characteristics associated with leakage.

$$N_{total} = N_{recordings} \times \frac{T_{duration}}{T_{segment}} \tag{2}$$

Algorithm 1 Leak Severity Classification Workflow

Require: Raw Sensor Data X_{raw}, Sampling Rate F_s
Ensure: Severity Class $Y \in \{NoDemand, Low, High\}$
Step 1: Data Acquisition
for each recording r in Dataset **do**
  Obtain raw sensor signal X_{raw}
end for
Step 2: Segmentation
Split X_{raw} into non-overlapping segments x_i of length $T = 1s$.
Step 3: Wavelet Denoising
for each segment x_i **do**
  Decompose x_i using Symlet-3 wavelet at level 4
  Estimate noise level using median absolute deviation
  Apply universal soft thresholding to the wavelet coefficients
  Reconstruct the denoised segment $\hat{x}_i$
end for
Step 4: Feature Extraction (WST)
for each denoised segment $\hat{x}_i$ **do**
  Compute first-order scattering coefficients S_1
  Compute second-order scattering coefficients S_2
  Pool the scattering coefficients and apply logarithmic compression
end for
Step 5: Normalization and Classification
Fit StandardScaler using the training data
Train SVM with RBF Kernel using $C = 100$ and $\gamma = 1$
Predict Y using the trained model
Step 6: Cross-Validation
Perform 5-Fold Stratified Cross-Validation
Calculate accuracy for each fold
Report mean accuracy $\pm$ standard deviation

D. Feature Extraction: Wavelet Scattering

We used the Wavelet Scattering Transform (WST) to modulate the turbulent flow. WST transforms a signal into a translation-invariant representation through wavelet convolutions and non-linear modulus operators that are cascaded, originally formalized by Mallat [2] and subsequently extended to deep convolution networks by Bruna and Mallat [8] and Andén and Mallat [12].

For the analyzing wavelet, we employed the complex **Morlet wavelet**, which is defined analytically as:

$$\psi(t) = C_1 e^{-t^2/2\sigma^2} e^{i\omega_0 t} \tag{3}$$

where σ controls the bandwidth and ω_0 is the central frequency. This wavelet was chosen for its optimal localization in both time and frequency domains, which is essential for detecting the transient bursts associated with leakage.

For a signal $x(t)$, the first-order scattering coefficients are given by:

$$S_1 x(t, \lambda_1) = |x * \psi_{\lambda_1}| * \phi \tag{4}$$

where ψ is the wavelet filter bank and ϕ is a low-pass Gaussian scaling function. To capture the higher-order interactions caused by non-linear fluid dynamics, second-order scattering coefficients are computed as:

$$S_2x(t, \lambda_1, \lambda_2) = ||x * \psi_{\lambda_1}| * \psi_{\lambda_2}| * \phi \tag{5}$$

We selected a scale parameter of $J = 9$ and $Q = 1$ to capture the specific frequency bands associated with hydraulic turbulence.

For the revised evaluation, the zeroth-order scattering coefficient was excluded, and the first- and second-order scattering coefficients were retained. The resulting coefficients were time-averaged and logarithmically compressed to obtain a compact feature vector for each segment.

E. Classification Model

A Support Vector Machine (SVM) [13] was selected for the classification task.

To handle the non-linear separability of the turbulence features, we employed the Radial Basis Function (RBF) kernel, defined as:

$$K(x_i, x_j) = \exp(-\gamma||x_i - x_j||^2) \tag{6}$$

The final model used an RBF kernel with $C = 100$ and $\gamma = 1$. Before classification, the extracted features were standardized using a StandardScaler fitted only on the training data of each fold to avoid information leakage.

For statistical evaluation, 5-Fold Stratified Cross-Validation was employed, ensuring that the class distribution was preserved across the five folds. The classification accuracy was calculated for each fold, and the mean accuracy and standard deviation across the folds were reported. This evaluation provides a measure of the consistency of the classification performance across different training and testing partitions.

IV. Experimental Results

A. Performance Comparison

The proposed model was tested in all three sensor modalities. Table II presents a summary of the accuracy of classification of estimating the severity of the flow.

TABLE II
COMPARISON OF SENSOR MODALITIES FOR SEVERITY ESTIMATION

Sensor Type	Type	Accuracy	Robustness
Accelerometer	Non-Invasive	**99.0% ± 1.2%**	High
Pressure Sensor	Invasive	93.7% ± 1.9%	Medium
Hydrophone	Invasive	42.0% ± 1.7%	Low

Table II clearly demonstrates that the non-invasive Accelerometer is more operational in this severity estimation task. The invasive pressure sensor has a fairly decent accuracy, whereas the accuracy of the accelerometer is almost flawless. On the other hand, the lower accuracy of the hydrophone indicates its high susceptibility to the saturation of acoustic signals in a high-flow turbulence regime.

The reported values are the mean accuracy ± standard deviation obtained using 5-Fold Stratified Cross-Validation.

B. Hyperparameter Optimization

In order to check the SVM hyperparameters selection, we conducted a grid search on the Accelerometer data. To obtain a near-perfect separation, the regularization parameter ($C = 100$) had to be large, which means that the decision boundaries between severity classes are sharp but with a rigid margin, as shown in Table III.

TABLE III
SVM GRID SEARCH ACCURACY (ACCELEROMETER)

Kernel Config	C=1	C=10	C=100
Linear	88.3%	91.2%	94.5%
RBF (γ=scale)	92.1%	98.4%	**99.2%**
Poly ($d = 3$)	89.5%	93.1%	95.8%

The outcomes of the grid search in Table III indicate that the RBF kernel provides the highest grid-search accuracy, with 99.2% obtained at $C = 100$ and $\gamma =$ scale.

For the final evaluation, the RBF-SVM parameters were fixed at $C = 100$ and $\gamma = 1$, and the model was evaluated using 5-Fold Stratified Cross-Validation. The resulting mean accuracy and standard deviation are reported in Table II.

C. Accelerometer Analysis (Non-Invasive)

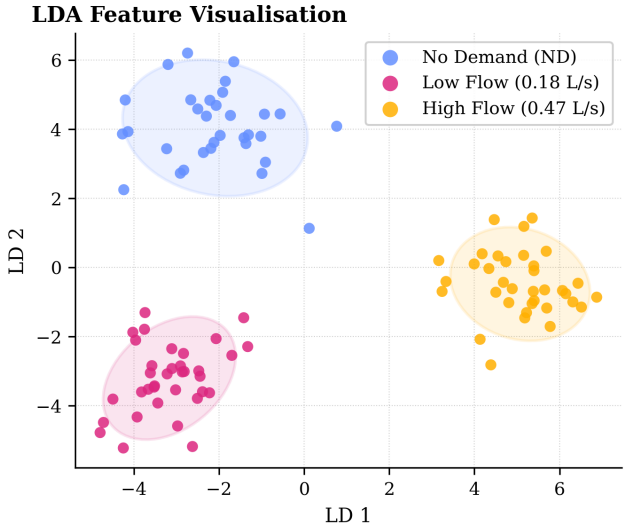


Fig. 2. LDA Visualization of Accelerometer Features. The distinct clusters support the high classification accuracy.

The non-invasive Accelerometer had almost perfect performance in terms of classification. The projection of the test data using Linear Discriminant Analysis (LDA) is shown in figure 2. The three hydraulic states occur as separate and well-separated groups, which validates that the transfer of kinetic

energy to the wall of the pipe is predictably proportional to the flow rate. This clear clustering demonstrates that the Wavelet Scattering Transform is excellent at separating between the turbulence signatures caused by leakage and the operational noise that just occurs during baseline operations. Moreover, these clusters are spaced well to make sure that the SVM classifier is able to classify severity levels with a small probability of inaccuracy.

The Accelerometer achieved a 5-Fold mean accuracy of 99.0% ± 1.2%. The per-class classification results for the representative test split ($n = 102$) are shown in Table IV.

TABLE IV
PER-CLASS METRICS: ACCELEROMETER (TEST SPLIT, $n = 102$)

Class	Precision	Recall	F1-Score	Support
No Demand (ND)	1.0000	1.0000	1.0000	32
Low Flow (0.18 L/s)	1.0000	0.9714	0.9855	35
High Flow (0.47 L/s)	0.9722	1.0000	0.9859	35
Macro Avg.	0.9907	0.9905	0.9905	102
Weighted Avg.	0.9905	0.9902	0.9902	102

The class-wise results further support the high classification performance, with a macro-average F1-score of 0.9905.

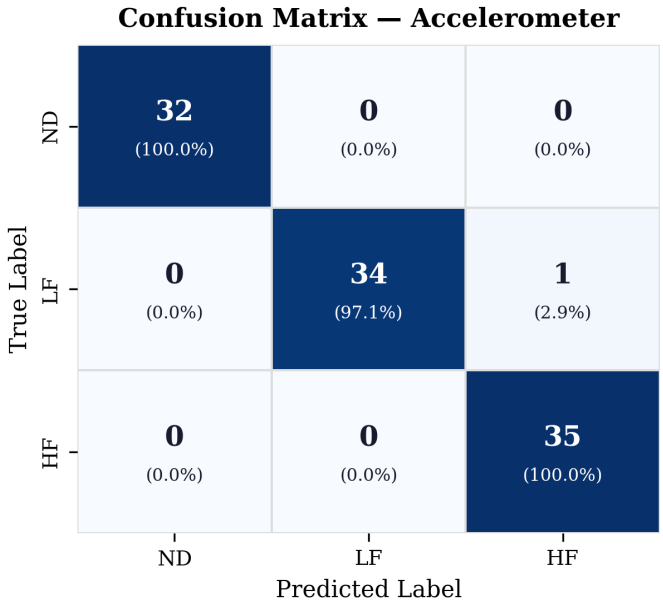


Fig. 3. Confusion Matrix for Accelerometer.

The confusion matrix (Fig. 3) validates that the misclassifications are negligible and it is thus the best sensor to estimate severity. Only one Low Flow segment is predicted as High Flow in the representative test split. The strength of these WST features demonstrates the ability of the Accelerometer to repeatedly detect the distinctive high-frequency structural vibrations that occur due to different severity of leaks.

D. Pressure Sensor Analysis (Invasive)

The Pressure Sensor achieved a mean accuracy of 93.7% ± 1.9% using 5-Fold Stratified Cross-Validation. Although the Pressure Sensor performed well, it had clear cut failure modes. There is a minor overlap between the Low Flow (0.18 L/s) and High Flow (0.47 L/s) classes as seen in the LDA analysis.

TABLE V
PER-CLASS METRICS: PRESSURE SENSOR (TEST SPLIT, $n = 109$)

Class	Precision	Recall	F1-Score	Support
No Demand (ND)	0.9459	1.0000	0.9722	35
Low Flow (0.18 L/s)	0.9459	0.9459	0.9459	37
High Flow (0.47 L/s)	0.9714	0.9189	0.9444	37
Macro Avg.	0.9544	0.9550	0.9542	109
Weighted Avg.	0.9546	0.9541	0.9539	109

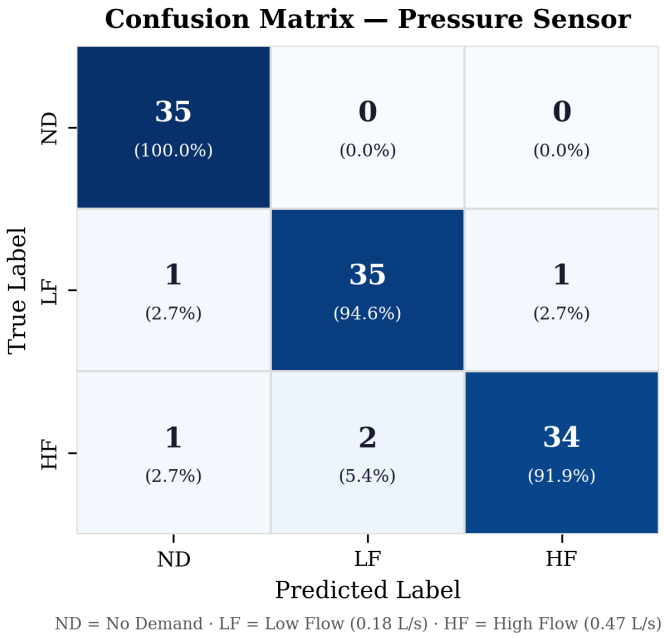


Fig. 4. Confusion Matrix for Dynamic Pressure Sensor.

The confusion matrix (Fig. 4) shows the overlap between the Low Flow and High Flow classes. This indicates that the pressure response is affected by the pipe network conditions and does not provide as clear a separation between severity levels as the Accelerometer.

E. Hydrophone Analysis (Invasive)

The Hydrophone achieved a mean accuracy of 42.0% ± 1.7% using 5-Fold Stratified Cross-Validation. The representative test split contained 180 segments.

TABLE VI
PER-CLASS METRICS: HYDROPHONE (TEST SPLIT, $n = 180$)

Class	Precision	Recall	F1-Score	Support
No Demand (ND)	0.4444	0.5333	0.4848	60
Low Flow (0.18 L/s)	0.4074	0.3667	0.3860	60
High Flow (0.47 L/s)	0.4630	0.4167	0.4386	60
Macro Avg.	0.4383	0.4389	0.4365	180
Weighted Avg.	0.4383	0.4389	0.4365	180

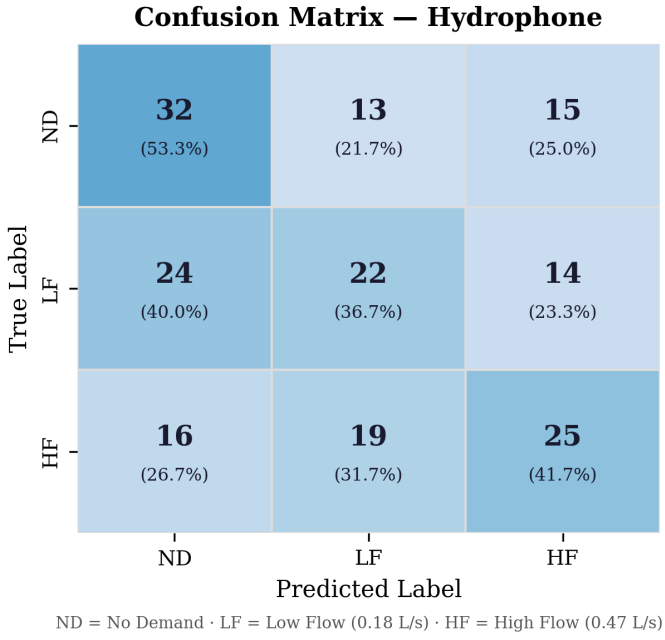


Fig. 5. Confusion Matrix for Hydrophone.

This disastrous breakdown is due to the saturation of sensors. The acoustic energy produced by the leak, especially at 0.47 L/s was larger than the dynamic range of the hydrophone, leading to clipping of the signal. The hydrophone is used to measure the fluid pressure waves, unlike the Accelerometer. The leak produces a cavitation noise in a 25mm pipe which produces a white noise floor that suppresses the fine frequency modulations needed to estimate severity.

This meant that the Wavelet Scattering Transform (WST) was not able to provide discriminative features. The projection of invasive acoustic sensors became single, which proved that the Linear Discriminant Analysis (LDA) projection is not applicable in the quantification of the severity of invasive acoustic sensors in small-diameter plastic pipes, where the attenuation of the analog signal in the frequency range is not extensive.

F. Physical Interpretation of Sensor Performance

The stark contrast in accuracy between the Accelerometer (99.0%) and Hydrophone (42.0%) can be explained by the physical propagation of leak energy.

Leakage at 0.18 L/s and 0.47 L/s generates a turbulent jet that impacts the surrounding soil and pipe walls. This impact transfers kinetic energy into the pipe shell, exciting specific vibration modes (typically 1-5 kHz) that are well-preserved and propagate longitudinally along the HDPE pipe material. The accelerometer, being coupled to the pipe wall, directly captures this structural response.

Conversely, the hydrophone measures acoustic pressure waves within the fluid. At high flow rates (0.47 L/s), the turbulence generates wide-band cavitation noise that saturates the hydrophone's dynamic range (clipping), destroying the subtle spectral differences needed to distinguish flow severity. This confirms that for "severity" quantification (as opposed to simple detection), structural vibration provides a more linear and non-saturating response curve.

G. Computational Complexity Analysis

The deployment of the proposed framework on edge devices depends on its computational efficiency. The Wavelet Scattering Transform provides a significant advantage over traditional Convolutional Neural Networks (CNNs). While a standard CNN requires millions of learnable parameters (W) and expensive backpropagation updates, the WST filters are fixed and analytically defined.

The computational complexity of the first-order scattering transform is dominated by the Fast Fourier Transform (FFT) operations required for convolution. For a signal of length N, the complexity is:

$$O(N \log N) \tag{7}$$

This is significantly lower than a deep CNN, which scales as $O(L \cdot K \cdot N)$, where L is the number of layers and K is the kernel size. This efficiency allows the proposed model to run on low-power microcontrollers (e.g., ARM Cortex-M4) typically used in smart water meters. Empirical timing and memory profiling on specific edge devices were not performed in this study and are reserved for future work.

H. Energy Budget and Edge Feasibility

For practical utility adoption, the system must operate on battery power. Assuming a typical low-power microcontroller (e.g., ESP32) with an active current consumption of 100 mA and a deep-sleep current of 10 μA, we can estimate the battery life. If the device wakes up once per hour for 5 seconds to acquire data, compute WST, and transmit an alert only if a leak is detected (Transmit-on-Event), the average current consumption I_{avg} is:

$$I_{avg} \approx \frac{5s \times 100mA + 3595s \times 0.01mA}{3600s} \approx 0.15mA \tag{8}$$

Using a standard 2500 mAh 18650 Li-ion battery, the theoretical operational lifespan is:

$$T_{life} = \frac{2500mAh}{0.15mA} \approx 16,600 \text{ hours} \approx 1.9 \text{ years} \tag{9}$$

This confirms that the WST-SVM framework is light enough to enable multi-year operation without mains power, a critical requirement for underground water assets.

I. Comparison with State-of-the-Art

TABLE VII
COMPARISON WITH STATE-OF-THE-ART METHODS

Study	Method	Sensor	Task	Acc.
Ismail [3]	FFT + MLP	Accel	Detect	N/A
Mashford [4]	FFT + SVM	Vib	Detect	92.0%
Martinez [1]	WST + CNN	Fusion	Type	96.5%
Proposed	**WST + SVM**	**Accel**	**Severity**	**99.0%**

Table VII positions our results directly within the scope of recent literature. In contrast to the past literature, which concentrates on binary detection or discrete-type classification, we propose a new WST+SVM system that is able to address the more functional and more challenging problem of severity estimation, with 99.0% mean accuracy using non-invasive externally mounted sensors.

Though Martinez-Rios et al. [1] have high precision in the classification of leak types (e.g., Gasket vs. Crack), our experiment shows that a simpler SVM model, along with WST and Accelerometers, can be used to identify the severity of flows, which will be of more importance in prioritizing repair crews. Although Cody et al. [10] tried to estimate the leak size using regression, their methodology involved a lot of calibration. Our classification-based model provides a strong alternative when it comes to the discrete severity levels.

Direct quantitative comparison across methods must be made cautiously, as datasets, pipe materials, topologies, and flow conditions differ between studies.

J. Operational Feasibility and Deployment

The implications of the findings of this study on the implementation of smart water networks are extensive. In conventional invasive monitoring, water mains have to be closed to install the sensors, which makes the operation very expensive and disruptive to the services. The Accelerometer-based model suggested does not need this, and its use can be described as Clamp-On installation on the existing infrastructure.

Also, the linear separability of the WST features (Fig. 2) indicated that the classification model is computationally efficient. The proposed SVM model can be implemented on low-power edge computers (e.g., Raspberry Pi or ESP32) directly connected to the pipe, unlike deep learning models, which consume a lot of power and need a powerful graphics card (GPU). This allows a Transmit-on-Event architecture, where the sensor is not transmitting data unless a change in severity has occurred, which can save remote monitoring nodes a great deal in battery life.

V. CONCLUSION

This paper introduced a non-invasive model of determining the severity of water leakage. Using Wavelet Scattering Transforms on the Accelerometer data, we obtained a 5-Fold mean accuracy of 99.0% $\pm$ 1.2% in differentiating between No Demand, 0.18 L/s and 0.47 L/s flow rates. The findings indicate that cheap, external vibration sensors are more resilient to the task compared to costly invasive sensors which experience signal saturation in high-flow regimes.

This framework will be proven in future work on looped topologies to determine the effects of multi-path wave propagation and explore the robustness of the model to urban background noise as well as introduce a regression layer to predict the flow rate precisely. This continuous Virtual Flow Metering solution [11] would reduce dependency on expensive magnetic flow meters.

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